Course & Code: B.E. (CSE) — Major Project Phase-II (BCS786)	**Project Guide:** Prof. KALAISELVI S
Project Title: SMART CLASSROOM MONITORING SYSTEM (SmartCampus AI)	**Team Leader:** MANJUNATHA R (1VI23CS058)
Team Members: A ABHIJITH (1VI23CS001), ANVIK C SHETTY (1VI23CS012)	**Document:** Project Work Progress Diary / Logbook

Weeks	Work Done / Progress	Discussion with Guide	Results / Decisions	Signature / Remarks
Week 1 20/07/2026 to 25/07/2026	**Main Activity:** System architecture design, backend setup and database integration. Initialized Turborepo monorepo, authored relational PostgreSQL schema with Prisma/SQLAlchemy, and established Redis caching layer.	**Discussed:** Designing a decoupled microservices architecture and choosing the right caching strategy for real-time WebSocket communication and state management.	**Decided:** Adopted Redis for Pub-Sub and PostgreSQL with Prisma/SQLAlchemy for relational data models. **Next Action:** Begin work on the core machine learning and face recognition pipeline.	*Guide Sign & Date:* *Remarks:*
Week 2 27/07/2026 to 31/07/2026	**Main Activity:** Implemented face recognition engine (InsightFace SCRFD 10G) and FAISS indexing for fast similarity search with 512D ArcFace embeddings.	**Discussed:** How to handle video processing efficiently and deduplicate recognized faces to prevent spamming the database during live video streams.	**Decided:** Implemented a clustering approach for extraction, 3-frame temporal persistence logic, and fine-tuned confidence thresholds. **Next Action:** Connect the backend AI pipeline to a user-facing dashboard.	*Guide Sign & Date:* *Remarks:*
Week 3 03/08/2026 to 08/08/2026	**Main Activity:** Built React frontend dashboard, integrated real-time analytics, implemented back-row 1.8x ROI small-face upscaling, MediaPipe pose quality filtering, and fully Dockerized the system.	**Discussed:** Handling distant back-row student faces, filtering bad head pose angles (-30° to +30°), and ensuring the system is easy to run across different environments.	**Decided:** Created docker-compose.yml and setup automation for one-click deployment. Added MediaPipe yaw/pitch filtering. **Next Action:** Enhance multi-student tracking to handle long occlusions without ID swaps.	*Guide Sign & Date:* *Remarks:*

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Week 4 10/08/2026 to 15/08/2026	**Main Activity:** Implemented Zero-Regression Spatial IoU Temporal Multi-Face Tracker (face_tracker.py) with velocity estimation, anti-switching logic, and rolling representation buffer.	**Discussed:** Addressing identity switching when closely seated students interact and preventing lost tracks when students look down to take notes.	**Decided:** Configured 90-frame decay buffer (45s at 2 FPS) for lost track recovery, motion trajectory gating, and retained top 15 highest-quality crops. **Next Action:** Build enterprise NestJS core services for rosters and attendance logging.	*Guide Sign & Date:* *Remarks:*
Week 5 17/08/2026 to 22/08/2026	**Main Activity:** Developed NestJS core services including Student Rosters, CSV/Excel batch ingestion, Timetable scheduling, attendance risk profiling, and MQTT camera heartbeat monitor.	**Discussed:** Structuring institutional academic hierarchy (Department → Program → Semester → Section) and automating chronic absenteeism risk alerts.	**Decided:** Implemented 4-tier risk classification (Critical, High, Medium, Safe), bulk database upserts using Prisma createMany, and integrated Mosquitto MQTT. **Next Action:** Build responsive administrative web dashboard with live RTSP video feeds.	*Guide Sign & Date:* *Remarks:*
Week 6 24/08/2026 to 29/08/2026	**Main Activity:** Built Next.js 16 dark-mode glassmorphic web dashboard, RTSP live multi-camera monitoring hub with HTML5 canvas bounding box overlays, and real-time WebSocket sync.	**Discussed:** Reducing UI rendering latency during multi-camera live video feeds and providing faculty with intuitive attendance statistics.	**Decided:** Replaced state-driven bounding box DOM re-renders with 60 FPS HTML5 canvas overlays, added auto-reconnecting WebSockets, and created departmental analytics. **Next Action:** Build contactless classroom video face enrollment studio and solve complex crop image delivery.	*Guide Sign & Date:* *Remarks:*

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Week 7 31/08/2026 to 05/09/2026	**Main Activity:** Created Classroom Face Enrollment Studio, designed two-stage decoupled processing pipeline (tracking first, deferred embedding extraction), and built safe crop URL resolver.	**Discussed:** Minimizing server memory exhaustion during full classroom video processing and handling diverse storage image path formats without frontend crashes.	**Decided:** Postponed heavy ArcFace inference until tracks confirm quality crops (saving 65% RAM), created polymorphic resolveCropUrl(), and added 4s AbortController timeouts. **Next Action:** Solve over-clustering where 25 classroom students produce 50+ fragmented candidate clusters.	*Guide Sign & Date:* *Remarks:*
Week 8 07/09/2026 to 12/09/2026	**Main Activity:** Implemented track-level ArcFace embedding centroid aggregation, precomputed cosine distance HDBSCAN clustering, and representative medoid selection.	**Discussed:** Eliminating fragmented duplicate clusters caused by clustering individual video frames rather than persistent student tracking trajectories.	**Decided:** Aggregated samples into L2-normalized track centroids, clustered tracks using precomputed cosine distance matrices, and selected medoids for FAISS indexing. Reduced clusters from 54+ to 28. **Next Action:** Formulate biometric data governance and build complete synchronization reset workflows.	*Guide Sign & Date:* *Remarks:*
Week 9 14/09/2026 to 19/09/2026	**Main Activity:** Engineered Biometric Data Governance & Synchronization Reset API, atomic FAISS index purge, disk metadata synchronization, and dual-confirmation safety modal in web UI.	**Discussed:** Student biometric data privacy compliance (GDPR/FERPA), preventing ghost recognitions from stale vector files, and protecting production records against accidental deletion.	**Decided:** Built synchronized PostgreSQL/FAISS atomic reset endpoints, implemented audit logging in EnrollmentHistory table, and added typed-confirmation administrative modals. **Next Action:** Formulate mathematical identity consolidation rules to stop similar-looking students from merging.	*Guide Sign & Date:* *Remarks:*

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Week 10 21/09/2026 to 26/09/2026	**Main Activity:** Formulated Multi-Evidence Safe Identity Consolidation Matrix with strict under-merging invariants, anti-chaining guards, and JSON merge decision telemetry logging.	**Discussed:** Preventing false identity merges between unrelated students having borderline similarity (0.70–0.76) and ensuring the system strictly adheres to: 1 Physical Person = 1 Candidate.	**Decided:** Enforced 7-point multi-evidence criteria (Centroid ≥ 0.82, Medoid ≥ 0.78, ≥3 pairs ≥ 0.80, Top-4 mean ≥ 0.82, min track sim ≥ 0.75, zero FAISS contradiction). System strictly prefers under-merging. **Next Action:** Validate cluster purity and implement automated multi-identity cluster splitting.	*Guide Sign & Date:* *Remarks:*
Week 11 28/09/2026 to 03/10/2026	**Main Activity:** Implemented connected-component graph analysis (Step 4.5), automated multi-identity cluster splitting via BFS, internal cluster purity scoring, and diagnostic UI badges.	**Discussed:** Detecting and self-healing accidental multi-student contamination inside a candidate cluster before final roster confirmation by faculty.	**Decided:** Constructed track centroid adjacency graphs at threshold 0.76, automatically partitioned disconnected components into pure sub-clusters, and added Evidence (Strong/Mod/Weak) badges. **Next Action:** Develop automated 12-test invariant verification test suite and complete final production build.	*Guide Sign & Date:* *Remarks:*
Week 12 05/10/2026 to 10/10/2026	**Main Activity:** Built 12-test automated unit test suite (test_identity_consolidation.py), ran 18 hybrid pipeline tests, completed Next.js 16 Turbopack production build, and finalized Phase-II documentation.	**Discussed:** Comprehensive system verification, testing boundary edge cases, final project demonstration readiness, and institutional report submission for Phase-II.	**Decided:** All 12 identity consolidation unit tests passed (100%), verified zero false merges across real classroom footage, and compiled full production artifacts for final evaluation. **Next Action:** Final project viva presentation, demonstration of live contactless attendance platform.	*Guide Sign & Date:* *Remarks:*